

TECHNICAL BRIEF • 28 AUGUST 2026

Arrhythmia classification, bit-exact on a RISC-V core

A 3,977-parameter CNN, quantized to 8.8 fixed point, running on the C-DAC VEGA ARIES v2 board — no accelerator, no cloud, 30 KB of activation RAM.

THEJAS32

VEGA ET1031 SoC

RV32IM

100 MHz · 256 KB SRAM

198 / 200

on real annotated beats

0.00 pp

quantization cost

Arrhythmias are episodic. Diagnosis is not.

A patient can be asymptomatic in clinic and symptomatic three hours later. Holter monitors record for offline review — they do not decide anything in the moment.

So the classifier has to live on the patient, and decide beat by beat.

What that constraint costs

Everything has to fit in 30 KB of activation RAM, run in integer arithmetic with no FPU, and finish inside the 800 ms a beat allows — while still matching what the model learned on a workstation. The rest of this deck is about whether it does.

1

Autonomy

A verdict that needs connectivity fails exactly when connectivity does.

2

Power

4,000 parameters on a low-power core is what makes multi-day monitoring plausible.

3

Privacy

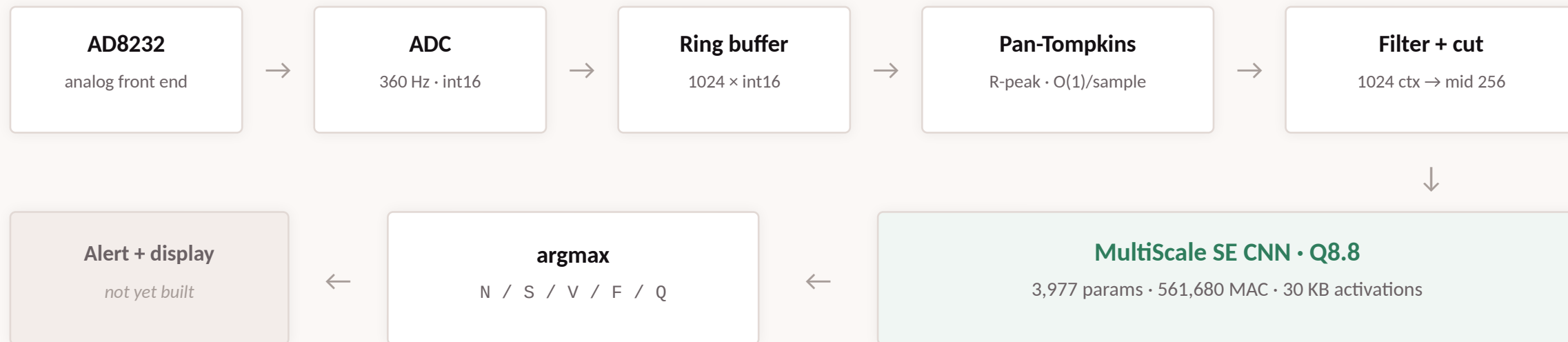
Raw ECG identifies a person. Transmit the verdict, not the waveform.

4

Determinism

A bare-metal loop has predictable timing. A phone sharing a CPU does not.

What actually happens to one heartbeat



Only the sampling path is hard real time — 2,778 μ s per sample at 360 Hz. Detection costs 13 μ s of that. Inference runs once per beat, triggered by a detected peak, in the slack between samples.

Small enough to fit, wide enough to see

Three parallel branches, different receptive fields.

A narrow kernel resolves the QRS upstroke. A wider one captures the overall width that marks a ventricular beat. The third is dilated — 13 taps of span for 7 taps of cost.

Added, not concatenated.

Fusing by addition keeps 8 channels entering the residual stack instead of 24 — the single largest contributor to the model's size.

Three details that silently break a port.

Branch 3 is dilated, not a plain 7-tap. The SE gate uses `hard_sigmoid` — `clip(0.2x+0.5,0,1)` — not a logistic sigmoid. Branches `Add()`, never concatenate.

```
input 256 x 1    (Q8.8, [-1,1])

|- Conv1D(8, k=3, d=1) -|
|- Conv1D(8, k=5, d=1) -|  BN>ReLU>SE
|- Conv1D(8, k=7, d=2) -|
    Add()

ResBlock1  8 -> 16, 1x1 shortcut
MaxPool1D(2)      256 -> 128
ResBlock2 16 -> 16, identity
MaxPool1D(2)      128 -> 64
GlobalAveragePooling1D -> 16
Dense(16) + ReLU
Dense(5)  -> argmax
```

No FPU. So the arithmetic changes.

Q8.8

± 128 range, 1/256 resolution

Safe because min-max normalization puts every input inside $[-1, 1]$ by construction. Measured worst-case weight error: 0.00195.

BN folded

into the preceding convolution

$\text{scale} = \gamma / \sqrt{\text{var} + \epsilon}$, $W' = W \cdot \text{scale}$, $b' = \beta + (b - \text{mean}) \cdot \text{scale}$. The kernel never runs a normalization layer. 3,625 values exported from 3,977 parameters.

int64 accumulate

requantize once per layer

Round-half-up $\gg 8$, then saturate to int16. An earlier HLS build needed ad-hoc per-layer shifts to avoid overflow; accumulating wide removes that class of bug.

The THEJAS32 implements RV32IM — integer multiply and divide, no floating-point unit. Every value in the inference path is a 16-bit integer.

METHOD

Eight gates, each checkable alone

Structured so a failure localizes to one layer instead of requiring a whole system to be debugged at once. Four defects were found this way — two of them essentially undiagnosable in an integrated system.

01

Toolchain and flashing

clean ELF loads and runs

02

CNN kernel on target

logits bit-identical

03

R-peak detector on target

peak indices bit-identical

04

Preprocessing parity

64.1% → 94.1% agreement

05

End-to-end ECG path

latency within 800 ms budget

06

Sensors, added singly

each plausible, ECG still runs

07

Fusion and local alerting

alert within one beat

08

Dashboard and rehearsal

ten minutes unattended

Bit-identical

Logits from the board matched the desktop reference digit for digit, beat 0 through beat 199. Peak indices matched sample for sample across five records.

Because the kernel is integer-only, this is binary. There is no tolerance to negotiate — any divergence would be a genuine portability defect, not rounding.

	beat 0	beat 199
desktop	1383 43 -1839	-493 -672 -704
VEGA board	1383 43 -1839	-493 -672 -704

198/200

float Keras
reference implementation

198/200

Q8.8 C kernel
on the board

0.00 pp

quantization cost
measured, not estimated

The detector, scored on real records

Record	Annotated	Detected	Se %	PPV %	Character
100	25	23	92.0	100.0	clean
106	21	20	95.2	100.0	
119	21	19	90.5	100.0	
208	33	24	72.7	100.0	frequent PVCs
222	25	18	72.0	100.0	variable atrial
excl. learning	112	102	91.1	100.0	0 false positives

Defect found here

The refractory check compared a detection-domain index against a refined one lying ~44 samples earlier. Every measured gap was inflated by the group delay.

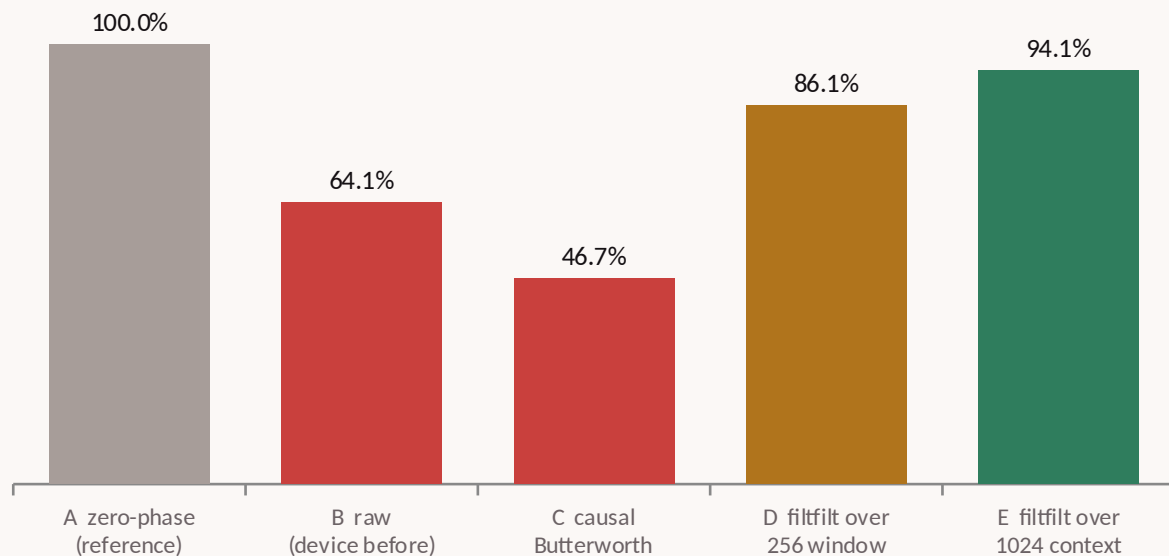
Result: reported beats 1, 2 and 7 samples apart. A heart cannot beat twice in 20 ms.

Fixing it removed all six false positives for 1.6 points of sensitivity. The right trade: a missed beat is silence, a false beat is a confident verdict on noise.

The detector is blind for its first two seconds while the adaptive threshold settles — that accounts for most of the shortfall on clean records. 208 and 222 are genuinely hard.

The gap nobody would have noticed

Training cuts beats from a zero-phase filtered signal — impossible on a device that sees one sample at a time. So the device cut from the raw signal. Was that fine? Measured, not assumed:



The inversion that gave it away

Raw beats sit CLOSER to the reference in waveform distance (0.075 vs 0.143) yet agree far LESS often. So distortion magnitude was not what changed the verdict.

Baseline wander, not distortion

Normalization is min-max over the window, so low-frequency drift rescales the whole beat. A 256-sample window cannot remove 0.5 Hz — one cycle is 720 samples. Filter 1024, cut the middle 256.

Per class the adopted fix is stronger than its average: SVEB 100%, VEB 99.5%, Fusion 100%. Cost on device: 4 KB of scratch and one forward-backward pass per beat.

Two accuracy figures, two questions

99.00%

intra-patient

Random beat split. Answers: does quantization degrade the model? Answer: no. Optimistic as a estimate for a new patient, because a person's beats appear on both sides.

81.1%

VEB, inter-patient

AAMI DS1/DS2 — 22 train patients, 22 disjoint test patients. The honest estimate for someone unseen. de Chazal 2004 reports 77.7% on the same split.

Class	Support	Se %	PPV %	Assessment
Normal (N)	44,240	68.1	93.6	
SVEB (S)	1,837	5.4	1.7	not usable
VEB (V)	3,220	81.1	37.3	beats the published reference
Fusion (F)	388	1.3	0.1	not usable

Why SVEB fails is architectural, not a training deficiency: it is defined by TIMING — early arrival, short preceding R–R interval — while its morphology closely resembles a normal beat. This model sees a single 256-sample morphology window and has no R–R input at all. 86% of SVEB beats are called Normal: systematic, not scattered.

LIMITATIONS

What this system does not do

No out-of-distribution rejection

The output is a five-way softmax, so it always produces one of five labels. An EEG trace, or a disconnected electrode, is classified as an arrhythmia with apparent confidence.

No rate reporting

Heart rate is implicit in peak spacing but is not computed or surfaced. Bradycardia and tachycardia are not reported.

Single verdicts are not clinical

Occasional ectopic beats occur in healthy people. What matters is burden over time, which is not yet aggregated.

No rhythm analysis

Classification is per beat and independent. Atrial fibrillation is defined by R-R irregularity across a strip, not by any one beat's shape — it is outside what this detects.

No per-patient calibration

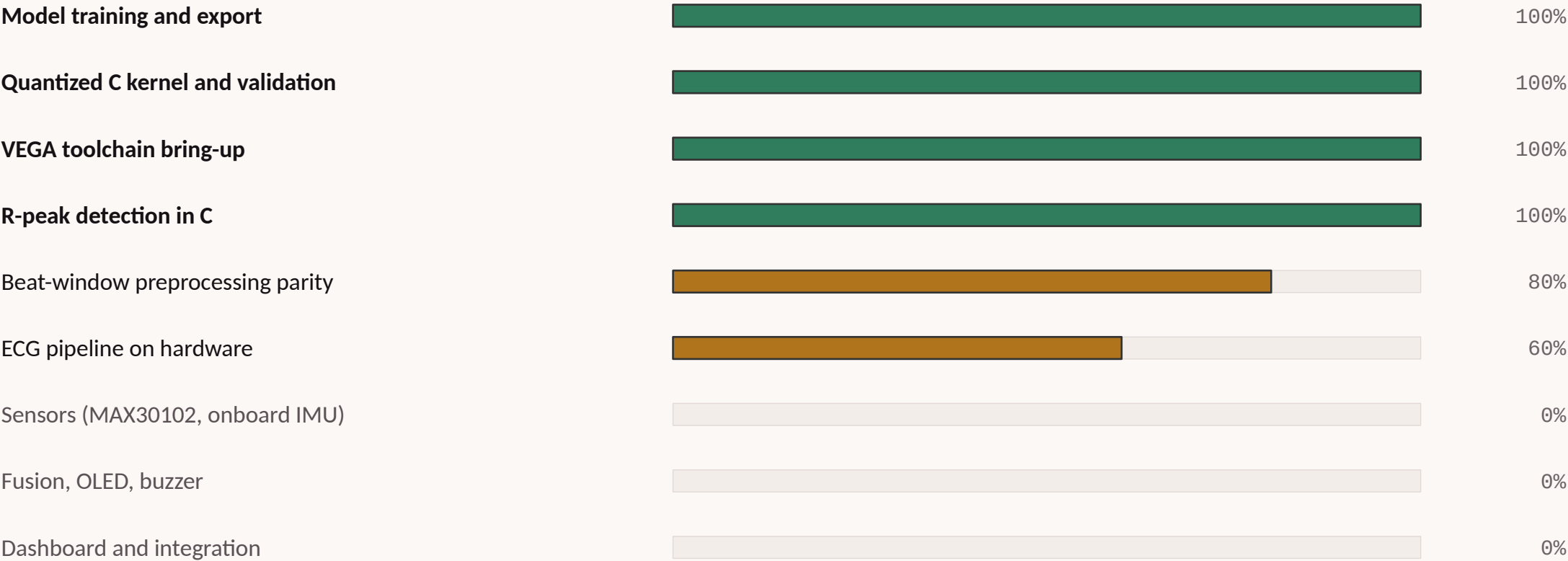
One fixed weight set, trained on a population, applied to everyone. No adaptation to an individual's baseline.

Detector degrades on hard records

72% sensitivity where ectopy is frequent or atrial morphology varies, against 92–95% on clean records.

Stated plainly because each constrains how the output may be read — and because a limitation you name yourself is a limitation you understand.

Where the work stands



68% complete by effort

The front-loaded portion is the part that could have failed in ways that invalidate everything downstream. None of it did.

WHAT THIS DEMONSTRATES

A 4,000-parameter classifier, matching its float model exactly, on a 100 MHz RISC-V core

The contribution is the verification chain

Each layer of the port was made independently checkable, which is what let four defects be localized precisely rather than inferred.

Two would have been near-undiagnosable integrated

A refractory period compared across mismatched index domains; a fixed-point filter losing its signal to truncation at poles of 0.993.

The preprocessing finding generalizes

A train/deploy conditioning gap produces no error and no warning. Here it silently altered 36% of verdicts, and was found only by measuring it.

Generalization is reported unflattered where the evidence requires it — 81.1% VEB sensitivity on unseen patients is real; SVEB and fusion detection are not usable, for architectural reasons named rather than obscured behind an aggregate figure.